

# Notes: Darmouni & Sutherland (JFE, 2024)

## Investment when new capital is hard to find

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# 1 Big picture

- **Question.** How does a fixed-supply shortage for new capital goods affect investment, secondary markets, and capital reallocation across firms?
- **Empirical shock.** The paper studies pandemic-era production disruptions in 2021–2022, especially the post-November 2021 period, when new equipment became harder to obtain.
- **Data advantage.** The authors use UCC filings to observe the financing and leasing of physical equipment at the asset level. The data record collateral descriptions, borrower and lender names, locations, manufacturers, model years, and serial numbers.
- **Main market mechanism.** When new equipment is scarce, some firms turn to used equipment. Used-equipment demand rises, assets trade more quickly, buyers search farther, matches occur across less similar firms, and prices rise.
- **Distributional mechanism.** Secondary markets partly absorb the shock for firms that can compete for used equipment. But they also transmit the shock to young firms, because older or more productive firms enter the used market and crowd out the younger firms that normally rely on used assets.
- **Main empirical conclusion.** Secondary markets are both shock absorbers and shock transmitters. They reallocate scarce capital toward some firms while imposing a distributive externality on others.

The paper is best read as a general-equilibrium investment paper rather than a narrow equipment paper. The novelty is that the authors observe the secondary-market channel directly. A new-capital shortage does not simply reduce aggregate investment one-for-one. Instead, it changes the relative scarcity of new and used assets. That relative scarcity changes who buys used assets, how far used assets move, and which firms end up investing less. This is why the results matter for capital allocation: the same shock can simultaneously raise trade in used capital, increase reallocation, and reduce investment for young firms.

**Economic message:** The central economics is a crowding mechanism. Used-capital markets make the economy more flexible, but flexibility is not neutral. When intermediate or older firms move into used markets, their willingness to pay changes the equilibrium faced by young firms. The paper therefore shows why a functioning resale market can soften a shock in aggregate while making the shock more unequal across firms.

## 2 Data and measurement

### 2.1 UCC equipment transaction data

The sample comes from UCC filings for secured lending and leasing on business equipment. Each observation records borrower, lender, borrower location, collateral descriptions, manufacturer, model, model year, and serial number. Serial numbers allow the authors to track the same physical asset across time, users, locations, and sectors.

Figure 2 is not just a data illustration. It is part of the identification logic because it explains how an administrative legal filing becomes an asset-level transaction panel. The ability to identify a collateral item by serial number is essential: without it, the paper could observe equipment financing but not secondary-market movement of the same asset. Table 1 then establishes that the dataset is large enough to study cross-asset, cross-sector, and cross-firm-age differences. The reported average equipment age and the large new-equipment share are important benchmarks. They imply that used markets are active even before the shock, but that new equipment remains a major component of investment.

**Economic message:** The data let the authors move from aggregate investment to the mechanics of re-allocation. The paper can ask not only whether firms invest less, but whether the same bulldozer, truck, tractor, or equipment unit changes users faster, travels farther, or moves into a different sector.

**Reading cautions:** UCC filings do not capture all equipment purchases. They are strongest for financed or leased assets. The paper’s conclusions therefore speak most directly to the market for financed business equipment, especially private and smaller firms that rely on secured credit or leasing.

### 2.2 Equipment age, firm age, and secondary-market outcomes

A transaction is coded as new capital when the collateral is new equipment. For used equipment, equipment age is measured using model year and transaction date. Firm age comes from Dun & Bradstreet and proxies for financing constraints, life-cycle stage, and market access. Secondary-market activity is measured by time since last contract for the same asset, distance between users, same-county status, same-SIC2 status, and same modal equipment code.

Table 2 is the first direct evidence of substitution. If the shock were simply a decline in demand or a financing contraction, one might expect fewer transactions of all types. Instead, the results show movement away from new capital and toward older equipment. Table 3 shows that this substitution is not just a passive compositional change. The secondary market becomes tighter: assets turn over more quickly, matching becomes less local, and buyers are willing to transact with sellers that are less similar by sector or equipment type. Figure 3 adds the firm-life-cycle dimension. Normally young firms use older capital, but the shock changes who competes for those assets.

**Economic message:** These measures capture the market-clearing adjustment. Quantity shifts show that firms substitute toward used assets. Trading-speed and distance measures show that search becomes more intense. Similarity measures show that firms broaden the set of acceptable counterparties. Together these outcomes imply that the shock affects the entire matching function in the market for used capital.

**Reading cautions:** Equipment age is not identical to capital quality. Older equipment can still be productive, especially if maintained well. The paper’s interpretation relies on the change in age, trading speed, and matching patterns together rather than age alone.

## 2.3 John Deere exposure and price data

The John Deere strike provides an additional source of variation in supply shortage exposure. Exposure is measured by Deere’s pre-period market share in an equipment category. The price analysis uses the subset of transactions for which actual transaction prices are available.

The Deere evidence is valuable because it sharpens the supply-side interpretation. The aggregate pandemic shock could be confounded by macro demand, interest rates, or credit conditions. The Deere strike is narrower: segments with higher Deere market share should be more exposed to the strike-induced shortage. Table 4 confirms the first-stage idea: new equipment availability falls more in exposed categories. Table 5 shows that these exposed categories also display greater secondary-market intensity. Table 6 shows that price effects move in the same direction, which is crucial because quantity substitution alone could reflect demand shifts. Prices rising alongside secondary-market activity is much more consistent with a shortage.

**Economic message:** The Deere tests show that the secondary-market response is not an artifact of calendar timing. When a specific producer disruption makes new equipment harder to obtain, used-equipment markets adjust through faster turnover, broader matching, and higher prices.

**Reading cautions:** The strike variation is not a randomized experiment at the firm level. It is an exposure design. Its strength comes from comparing equipment categories with different pre-existing Deere dependence while controlling for broader timing and category patterns.

## 2.4 Firm-age outcomes and reallocation measures

Firm age is central to the distributional results. The paper studies how used investment and total investment change by firm-age group after the shock. It also examines prior and current users of the same used capital to understand the direction of reallocation.

Table 7 shows the distributional externality in the used market. Young firms, which normally rely heavily on used equipment, do not benefit from the used-market boom. Intermediate firms are the marginal entrants into the used market. Table 8 is even more important because it rules out a simple substitution story for young firms. If young firms merely substituted from new to used or from used to new, total investment might be unchanged. Instead, total investment falls for them, indicating crowding out. Table 9 then describes the direction of asset movement: current users during the shortage are older and linked to sectors with stronger productivity growth than prior users.

**Economic message:** The shortage reallocates capital toward firms that can compete in the tightened used market. That may improve some allocative margins if assets move toward more productive-sector users, but it also limits entry, growth, and investment by young firms.

**Reading cautions:** Firm age is an informative proxy, but it is not a perfect measure of financial constraints or productivity. The paper therefore combines age-based results with user-comparison results rather than interpreting age alone as the full mechanism.

### 3 Theory and empirical specifications

#### 3.1 New versus used capital regression

$$NewCapital_{i,e,t} = \beta \mathbf{1}\{PostNov21\} + \nu_c + u_{e,t} + \zeta_m + \xi_i + \varepsilon_{i,e,t}.$$

A negative post coefficient for new-capital outcomes means that firms obtain fewer new assets after the shortage begins. A positive effect on equipment age means that the assets purchased are older. The strength of Table 2 is that it does not rely on aggregate investment alone. It distinguishes whether firms are buying a different type of capital. This distinction is central because the paper’s mechanism requires substitution from new to used equipment.

**Economic message:** The regression measures the first margin of adjustment: firms facing scarce new capital turn to older capital. This is the bridge from the supply shock to the secondary-market analysis.

#### 3.2 Trading activity regression

$$TradingActivity_{e,t} = \beta \mathbf{1}\{Year2021/22\} + FE + \varepsilon_{e,t}.$$

The dependent variable is time since the last contract for the same asset, geographic distance between users, same-county status, same-SIC2 status, or same modal equipment code.

The dependent variables are designed to translate market tightness into observable asset movements. Shorter time since last contract means assets are being recycled faster. Greater distance means buyers search beyond local markets. Lower same-county and same-industry matches mean the asset is being reallocated across broader boundaries. Lower same-equipment-code matching means buyers become more flexible about what assets can substitute for what they originally wanted.

**Economic message:** This specification shows that a secondary market can absorb part of a shock by increasing turnover and broadening matches. But these same forces can raise search costs and bid competition.

#### 3.3 John Deere strike design

$$Outcome_{e,t} = \beta(Strike_t \times DeereShare_{e,2019}) + Controls + \varepsilon_{e,t}.$$

The coefficient on the interaction term asks whether equipment categories more exposed to Deere experience stronger changes during the strike. Table 4 checks the first stage: new-equipment supply falls more in exposed categories. Table 5 checks the market response: used assets in exposed categories are traded faster and over broader market boundaries. Table 6 checks prices: price increases provide evidence that demand pressure and scarcity are raising equilibrium valuations. The three tables together form a coherent supply-shock test.

**Economic message:** The Deere design is the most persuasive mechanism evidence. It links a producer-specific disruption to both quantity and price responses in secondary markets.

#### 3.4 Firm-age investment regressions

$$UsedCapital_{e,a,s,t} = \sum_g \beta_g \mathbf{1}\{PostNov21\} \times AgeGroup_g + \nu_e + u_t + \varepsilon_{e,a,s,t},$$

$$TotalCapital_{e,a,s,t} = \sum_g \beta_g \mathbf{1}\{PostNov21\} \times AgeGroup_g + \nu_e + u_t + \varepsilon_{e,a,s,t}.$$

These regressions separate two possible interpretations. One interpretation is benign substitution: firms that cannot buy new equipment simply buy used equipment. Another interpretation is equilibrium crowding: some firms move into the used market and crowd out others. Table 7 shows that young firms are not the beneficiaries of the used-market expansion. Table 8 shows that their total investment declines, so the effect is not just a re-labeling of investment type. This makes the distributional externality visible.

**Economic message:** The firm-age regressions show that the secondary-market response is not distribution-neutral. In a shortage, the used market may allocate equipment toward firms with stronger ability to pay, leaving young firms with less capital.

### 3.5 Prior-user and current-user comparison

Table 9 compares prior users and current users for assets that move in the secondary market during the shock. The aim is to identify who gives up assets and who receives them.

Table 9 complements Tables 7–8. Those tables show investment losses by young firms; Table 9 asks where assets go. The fact that current users are older indicates that used capital is not simply circulating within the same young-firm segment. The productivity-growth comparison suggests that capital may move toward sectors where the marginal value of equipment is higher. But that does not make the distributional effect disappear: allocation to higher-productivity users can coexist with entry or growth constraints for young firms.

**Economic message:** Secondary markets reallocate scarce assets toward firms or sectors with higher willingness to pay. That can improve static allocation while harming young-firm investment and potentially future dynamism.

## 4 Identification, robustness, and interpretation

### 4.1 Evidence chain

The evidence chain combines time-series evidence around November 2021, cross-sectional exposure from the John Deere strike, price evidence, and firm-age investment regressions.

Figure 1 makes the shock plausible. Table 2 shows the direct substitution away from new equipment. Table 3 shows that the used market heats up. Tables 4–6 show that a sharper supply shock produces the same pattern in more exposed equipment categories. Tables 7–8 then reveal that young firms do not simply participate in the used-market expansion; they are crowded out. This sequence is important because no single table can prove the entire mechanism. The paper’s contribution is the consistency of the evidence across margins.

**Economic message:** The best way to read the paper is as an equilibrium test. The authors first identify the shock, then observe substitution, then observe market tightness, then observe who loses.

### 4.2 Main threats and how the tables address them

- **Generic macro demand.** If the results were only demand-driven, one would not expect the Deere-share exposure tests to line up with the same patterns.
- **Credit supply.** If the shock were only financial, used-equipment prices and exposed-category resale intensity would be harder to explain.
- **UCC sample selection.** Because the dataset covers financed equipment, cash purchases may be missing. The paper mitigates this by focusing on within-dataset movements and many fixed effects.
- **Young-firm demand collapse.** Tables 7–8 and the reallocation evidence help distinguish weak demand among young firms from competition that prices young firms out of used assets.

Price effects are essential because quantities alone can be ambiguous. Higher used-equipment prices in exposed markets indicate scarcity and competition. The age-distribution results then show that the price and matching changes have distributional consequences. If young firms simply had no demand for investment, one would not need the secondary-market mechanism. The paper’s point is that young firms lose when other firms’ demand for used capital rises.

**Economic message:** Robustness in this paper is not just about adding controls. It is about showing that different margins - prices, trading speed, geography, similarity, and firm-age investment - all move in the direction implied by the same mechanism.



## 5 Tables and figures with fully detailed interpretation

### 5.1 Figure 1: The 2021–2022 production disruptions

**Read:** The left panel shows equipment price inflation rising sharply relative to consumer price inflation. The right panel shows an increase in supply-chain pressure. Together, the figure places the empirical episode in a broader macro environment of capital-goods scarcity.

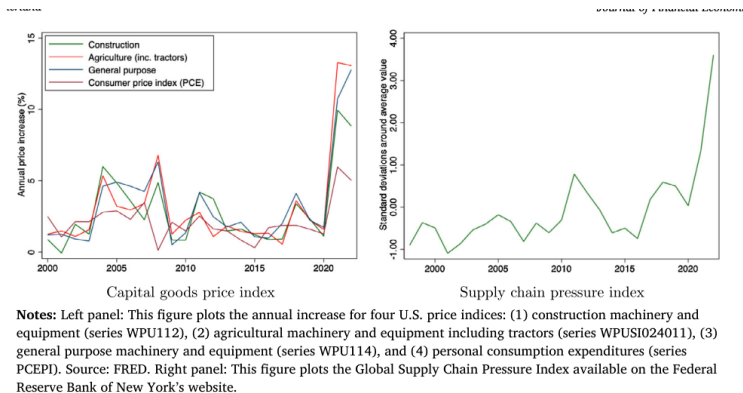


Fig. 1. The 2021–2022 Production Disruptions.

Figure 1: The 2021–2022 production disruptions

The purpose of Figure 1 is to establish that the paper is not studying an arbitrary calendar period. A fixed-capital supply shortage should show up in relative prices and supply-chain strain. The figure documents both. The price panel shows that equipment prices were not merely following general consumer-price inflation; the equipment market had its own scarcity episode. The supply-chain pressure panel gives a complementary nonprice indicator that production and delivery frictions were unusually severe.

**Mechanism and identification role:** Figure 1 is the starting point for the substitution mechanism. If new capital becomes hard to obtain, firms that still want to invest must either wait, reduce investment, or search for substitutes. Used capital is the natural substitute. Therefore, the figure motivates why Tables 2 and 3 should show substitution and secondary-market heat.

**Reading cautions:** Figure 1 alone is descriptive. It does not identify which firms or equipment types are most affected. The Deere-strike tests later in the paper provide more targeted supply-side evidence.

**Economic message:** The figure establishes the economic environment: new capital is scarce, so the market-clearing response must occur through prices, substitution, and reallocation.

### 5.2 Figure 2 and Table 1: UCC filings and summary statistics

**Read:** Figure 2 shows how the authors convert a financing filing into asset-level information. Table 1 reports the scale and baseline characteristics of the transaction sample.

## UCC-1 Form

### DEBTOR INFORMATION

Full name: LIEN SOLUTIONS

Email Contact at Filer: UCCFILINGRETURN@WOLTERSKLUEWER.COM

### SECURED PARTY INFORMATION

Contact name: LIEN SOLUTIONS

Mailing Address: P.O. BOX 29071

City, State Zip Country: GLENDALE, CA 91209-9071 USA

### COLLATERAL

Org. Name: SHAWMUT CORPORATION

Mailing Address: 208 MANLEY ST

City, State Zip Country: WEST BRIDGEWATER, MA 02379-1044 USA

### SECURED PARTY INFORMATION

Org. Name: TOYOTA INDUSTRIES COMMERCIAL FINANCE, INC.

Mailing Address: P.O. BOX 9050

City, State Zip Country: DALLAS, TX 75019-9050 USA

### SECURED PARTY INFORMATION

ALTERNATIVE DESIGNATION:

CUSTOMER REFERENCE: MA-0-67035011-56034684

### COLLATERAL

VE (1) TOYOTA FORKLIFT MODEL # 7FBEU18 SERIAL # 21898

TABLE 1  
Summary Statistics.

|                                | p25    | Median | p75    | Av.    | N   |
|--------------------------------|--------|--------|--------|--------|-----|
| Eq. Age (Y)                    | 0      | 1      | 6      | 4.77   | 12M |
| % New Eq.                      | -      | -      | -      | 60.5   | 18M |
| Used Eq. Estimated Value (USD) | 12,482 | 31,772 | 74,250 | 55,422 | 7M  |
| New Eq. Estimated Value (USD)  | 9,799  | 27,548 | 76,715 | 64,282 | 10M |
| Firm Age (Y)                   | 4      | 13     | 29     | 21.0   | 16M |

**Notes:** This table presents summary statistics for our transaction sample. We eliminate UCC filings in which the debtor is a wholesaler, equipment seller, rental or leasing company, auctioneer, or government. The time period includes contracts originated between January 1997 and March 2022. The estimated equipment values are provided by EDA. Firm age is from Dun & Bradstreet.

(a) Figure 2: Example UCC filing

(b) Table 1: Summary statistics

Figure 2 explains the data-generating process. The filing contains lender, borrower, collateral, and asset identifiers. The key variable is the serial number, which allows the same asset to be tracked across successive financing relationships. This is what makes the secondary-market analysis possible. Table 1 then tells the reader how broad the market is. The sample is not a handful of transactions; it includes a large set of equipment contracts. The average age of used assets and the large share of new-equipment transactions show that both primary and secondary markets are economically important.

**Mechanism and identification role:** The paper's mechanism depends on observing whether a given asset changes hands. Standard firm-level investment data cannot reveal that. The UCC data can reveal whether an asset trades faster, moves farther, or changes sectors. This is why Figure 2 and Table 1 are not merely background materials; they justify the empirical strategy.

**Reading cautions:** The data are strongest for secured and leased equipment. Cash purchases and unrecorded transfers are less visible. This matters for external validity, but the paper's within-UCC patterns remain informative for the financed equipment market.

**Economic message:** The paper's data allow capital reallocation to be measured directly, not inferred indirectly from firm-level investment rates.

## 5.3 Table 2: New versus used capital

**Read:** Table 2 shows that after the shortage period begins, firms are less likely to finance new equipment and the equipment they finance is older.

Table 2 is the first core result. It establishes the substitution margin. If new equipment is scarce, firms cannot always obtain the exact new assets they would like. The table shows that this scarcity changes observed transaction composition: new-equipment transactions fall and the age of equipment rises. This pattern supports the idea that firms substitute toward used capital rather than simply dropping out of investment markets completely.

**Mechanism and identification role:** Table 2 connects the macro shock to firm behavior. The table does not yet show whether secondary markets become tight or who is hurt, but it shows the first behavioral response required by the theory: buyers move from new to used assets.

**Reading cautions:** The coefficient should be interpreted as a change in transaction composition within the observed UCC market. It does not by itself measure the full quantity of all capital acquired in the economy.

**Economic message:** The table shows the immediate substitution effect of the shortage. Firms do not just invest less; they change the kind of capital they buy.

**Table 2**  
New vs. Used Capital Transactions.

|                    | (1)<br>New           | (2)<br>Log (1+Equipment Age) | (3)<br>New           | (4)<br>Log (1+Equipment Age) |
|--------------------|----------------------|------------------------------|----------------------|------------------------------|
| Post Nov 21 Crunch | -0.077***<br>[-9.02] | 0.163***<br>[6.68]           | -0.055***<br>[-7.80] | 0.098***<br>[6.90]           |
| Observations       | 2211339              | 1548352                      | 1887180              | 1280454                      |
| Adjusted $R^2$     | 0.532                | 0.694                        | 0.685                | 0.813                        |

$t$  statistics in brackets.

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

**Notes:** This table models equipment age as a function of time. The unit of observation is a transaction for a particular piece of equipment. The dependent variable in Columns 1 and 3 is an indicator variable equal to 1 if the equipment is new, and 0 if used; in Columns 2 and 4 the dependent variable is the log of equipment age. Columns 1 and 2 include county, manufacturer-model, and equipment code  $\times$  calendar month fixed effects, while Columns 3 and 4 add firm fixed effects. The sample includes all contracts from January 2019 to March 2022. Standard errors are clustered by equipment.

**Table 3**  
Trading Activity in Used Capital.

|                | (1)<br>Log Time Since Last Contract | (2)<br>Log Distance | (3)<br>Same County   | (4)<br>Same SIC2     | (5)<br>Same Modal Equipment Code |
|----------------|-------------------------------------|---------------------|----------------------|----------------------|----------------------------------|
| Year 2021/2    | -0.148***<br>[-11.99]               | 0.051**<br>[2.33]   | -0.018***<br>[-3.86] | -0.018***<br>[-3.00] | -0.011**<br>[-2.06]              |
| Observations   | 288517                              | 329559              | 345244               | 345244               | 345244                           |
| Adjusted $R^2$ | 0.443                               | 0.445               | 0.324                | 0.275                | 0.352                            |

$t$  statistics in brackets.

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

**Notes:** This table models secondary market activity as a function of time. The unit of observation is a transaction for a particular piece of equipment. The dependent variable in Column 1 is the log number of days between the last two transactions for that equipment; in Column 2 it is the log number of miles between the counties of the last two users of that equipment; in Column 3 it is an indicator variable for the last two users of that equipment being in the same county; in Column 4 it is an indicator variable for the last two users of that equipment being in the same two-digit SIC code; in Column 5 it is an indicator variable for the last two users of that equipment sharing the same modal equipment code defined over their previous transactions. All specifications include equipment serial number, county and calendar month fixed effects, as well as indicators for each year of equipment age. The sample includes all equipment that transacted at least twice. Standard errors are clustered by serial number.

## 5.4 Table 3: Used-capital trading activity

**Read:** Table 3 examines whether used assets trade faster and across broader market boundaries in 2021–2022.

Table 3 moves from substitution to secondary-market equilibrium. Shorter time since last contract means assets are being recycled more quickly. Larger distance means the buyer set expands geographically. A decline in same-county or same-SIC matches means buyers and sellers are less local and less similar. A decline in same modal-equipment-code matches suggests broader substitution across equipment categories. The table therefore captures a thicker and more strained used-capital market.

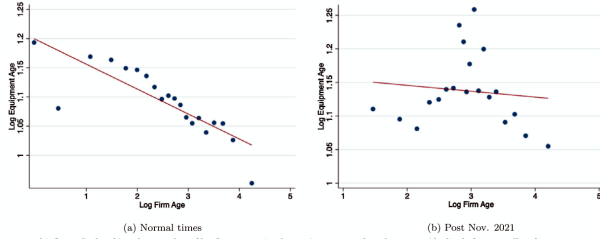
**Mechanism and identification role:** If used-capital demand increases, existing assets should be pulled from a wider set of sellers and locations. Table 3 is exactly this mechanism in the data. It is the market-clearing counterpart to Table 2’s transaction-composition result.

**Reading cautions:** These outcomes are proxies for search and matching intensity. They do not directly observe the search process itself. Their strength comes from moving together in the predicted direction.

**Economic message:** The used market does not passively absorb the shortage. It becomes more active, less local, and less specialized.

## 5.5 Figure 3 and Table 4: Firm-age gradient and Deere first stage

**Read:** Figure 3 shows the baseline relation between firm age and capital age, and how that relation changes during the shortage. Table 4 shows that equipment segments with greater pre-period Deere exposure experience less new equipment during the strike.



(a) Figure 3: Firm age and capital age

**Table 4**

John Deere Strike: Share of New Equipment.

|                | (1)<br>New Equipment | (2)<br>New Equipment |
|----------------|----------------------|----------------------|
| Deere x Strike | -0.081**<br>[-2.64]  | -0.077**<br>[-2.64]  |
| Observations   | 3055351              | 3055344              |
| Adjusted $R^2$ | 0.263                | 0.279                |

$t$  statistics in brackets.

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

**Notes:** This table models equipment age as a function of time and manufacturer. The unit of observation is a transaction for a particular piece of equipment. The dependent variable is an indicator variable equal to 1 if the equipment is new, and 0 if used. Deere is an indicator for equipment manufactured by John Deere. Strike is an indicator variable for transactions occurring between December 2021 and January 2022. Column 1 fixed effects: county, equipment code-month-year, manufacturer; Column 2 fixed effects: county, equipment code-month-year,

(b) Table 4: Deere strike and new equipment

Figure 3 is the bridge between market-level substitution and distributional incidence. In normal times, young firms tend to use older capital. This is intuitive: young firms have less collateral, less access to financing, and lower ability to buy new assets. When the shock causes older or intermediate firms to enter used markets, young firms face new competition in the very segment where they usually operate. Table 4 is the first-stage evidence for the Deere design. It shows that the strike made new equipment harder to obtain in categories that were more reliant on Deere.

**Mechanism and identification role:** Figure 3 explains why the shock can hurt young firms even if it is a new-equipment shortage. Table 4 strengthens the supply-shock interpretation by showing that a producer-specific disruption generates a new-equipment shortage in exposed segments.

**Reading cautions:** Figure 3 is descriptive, and Table 4 is an exposure design. Neither alone proves crowding out. The crowding-out result comes from combining these pieces with Tables 7–8.

**Economic message:** The shock reaches young firms indirectly. Young firms may not be major buyers of new capital, but they are exposed because other firms move into the used-capital segment.

## 5.6 Tables 5 and 6: Deere strike, used-market tightening, and prices

**Read:** Table 5 shows that high-Deere-share categories experience hotter used-capital trading during the strike. Table 6 shows price increases in exposed segments.

**Table 5**  
John Deere Strike: Trading Activity in Used Capital.

|                           | (1)<br>Log Time Since Last Contract | (2)<br>Log Distance | (3)<br>Same County   | (4)<br>Same SIC2  | (5)<br>Same Modal Equipment Code |
|---------------------------|-------------------------------------|---------------------|----------------------|-------------------|----------------------------------|
| Strike x Deere Share 2019 | -0.712***<br>[-4.62]                | 0.743***<br>[3.28]  | -0.163***<br>[-3.15] | -0.060<br>[-0.96] | -0.149**<br>[-2.48]              |
| Observations              | 165492                              | 197228              | 209986               | 209986            | 209986                           |
| Adjusted R <sup>2</sup>   | 0.447                               | 0.526               | 0.408                | 0.298             | 0.401                            |

t statistics in brackets.

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

**Notes:** This table models secondary market activity as a function of time and John Deere's market share in the equipment code. The unit of observation is a transaction for a particular piece of equipment. The dependent variable in Column 1 is the log number of days between the last two transactions for that equipment; in Column 2 it is the log number of miles between the counties of the last two users of that equipment; in Column 3 it is an indicator variable for the last two users of that equipment being in the same county; in Column 4 it is an indicator variable for the last two users of that equipment being in the same two-digit SIC code; in Column 5 it is an indicator variable for the last two users of that equipment sharing the same modal equipment code defined over the equipment's modal equipment code. The sample omits equipment manufactured by John Deere itself and includes only

(a) Table 5: Deere strike and trading activity

**Table 6**  
John Deere Strike: Prices for Other Manufacturers' Equipment.

|                           | (1)<br>Log Price-Used | (2)<br>Log Price-Used | (3)<br>Log Price-New | (4)<br>Log Price-New |
|---------------------------|-----------------------|-----------------------|----------------------|----------------------|
| Strike x Deere Share 2019 | 0.953**<br>[2.19]     | 1.015***<br>[3.11]    | 3.105***<br>[6.10]   | 3.145***<br>[4.67]   |
| Deere Share 2019          | 1.054<br>[1.42]       | 1.832*<br>[1.96]      | 1.170<br>[1.71]      | 2.061**<br>[2.40]    |
| Log (1+ Equipment Age)    | -0.602***<br>[-8.04]  | -0.639***<br>[-10.20] | -0.000<br>[-0.00]    | 0.009<br>[0.12]      |
| Observations              | 33666                 | 33666                 | 7037                 | 7037                 |
| Adjusted R <sup>2</sup>   | 0.325                 | 0.404                 | 0.479                | 0.562                |

t statistics in brackets.

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

**Notes:** This table studies equipment prices as a function of time and John Deere's market share in the equipment code. Strike is an indicator variable for December 2021 to January 2022. Deere Share 2019 is the share of contracts in that equipment code that involve equipment manufactured by John Deere, measured in 2019. The unit of observation is a transaction for a particular piece of equipment. Fixed effects in each column: month x year, state, and sector (odd columns); month x year, state, sector, and equipment family (even columns). The sample omits equipment manufactured by John Deere itself and includes only

(b) Table 6: Deere strike and prices

Tables 5 and 6 are mechanism tests. Table 5 shows that the exposure-driven shortage spills into the used market. Assets in exposed categories trade faster and across broader market boundaries. Table 6 adds the price response. This is critical because an increase in used transactions alone could reflect lower prices or weak demand. A price increase indicates scarcity and higher willingness to pay. The two tables together show that exposed markets tighten.

**Mechanism and identification role:** A supply shock to new equipment raises demand for used substitutes. The secondary market responds with higher turnover, broader search, and higher prices. Tables 5 and 6 verify all three pieces.

**Reading cautions:** Price data are available only for a subset of transactions. The price results should therefore be read with the trading-activity results, not in isolation.

**Economic message:** The Deere shock confirms the core mechanism in a sharper setting: when new equipment supply is disrupted, secondary markets become more valuable and more competitive.

## 5.7 Tables 7 and 8: Firm-age distributional effects

**Read:** Table 7 shows how used investment changes by firm age. Table 8 shows how total equipment investment changes by firm age.

**Table 7**  
Investment in Used Capital Across Firm Age Distribution.

|                                      | (1)<br>Log Used Capital | (2)<br>Log Used Capital |
|--------------------------------------|-------------------------|-------------------------|
| Post Nov 21 x Firm Age Below 4y      | -0.104**<br>[-2.69]     | -0.128***<br>[-3.29]    |
| Post Nov 21 x Firm Age Btw 4 and 29y | 0.016**<br>[2.08]       | 0.014<br>[1.50]         |
| Firm Age Below 4y                    | -0.162***<br>[-14.02]   |                         |
| Firm Age Btw 4 and 29y               | 0.294***<br>[77.56]     |                         |
| Observations                         | 318765                  | 318757                  |
| Adjusted $R^2$                       | 0.311                   | 0.375                   |

$t$  statistics in brackets.

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

**Notes:** This table models the log number of transactions for used equipment as a function of time and borrower age. Post Nov 21 is an indicator for months starting in November 2021. We measure borrower age using indicator variables for borrowers 3 years or younger, 4–29 years old, or at least 30 years old. The unit of observation is state-equipment code-firm age group-month-year. Fixed effects in each column: (1) equipment code and month  $\times$  year; (2) equipment code, age, and state  $\times$  month  $\times$  year. The sample includes all contracts for used equipment from January 2019 to February 2022. Standard errors are clustered by month-year.

(a) Table 7: Used investment by firm age

|                                      | (1)<br>Log Total Capital | (2)<br>Log Total Capital |
|--------------------------------------|--------------------------|--------------------------|
| Post Nov 21 x Firm Age Below 4y      | -0.170***<br>[-4.54]     | -0.205***<br>[-4.99]     |
| Post Nov 21 x Firm Age Btw 4 and 29y | 0.008<br>[1.54]          | 0.004<br>[0.70]          |
| Firm Age Below 4y                    | -0.356***<br>[-26.43]    |                          |
| Firm Age Btw 4 and 29y               | 0.277***<br>[72.10]      |                          |
| Observations                         | 318765                   | 318757                   |
| Adjusted $R^2$                       | 0.389                    | 0.470                    |

$t$  statistics in brackets.

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

**Notes:** This table models the log number of equipment transactions as a function of time and borrower age. Post Nov 21 is an indicator for months starting in November 2021. We measure borrower age using indicator variables for borrowers 3 years or younger, 4–29 years old, or at least 30 years old. The

(b) Table 8: Total investment by firm age

These are the paper’s most important distributional tables. Table 7 shows that young firms do not gain from the used-market expansion. Instead, they lose used investment relative to older groups, while intermediate firms are more likely to maintain or expand used investment. Table 8 then shows that young firms lose total investment as well. This is the key evidence against a simple substitution story. If young firms merely shifted to another type of capital, total investment would not fall. The decline in total investment indicates crowding out.

**Mechanism and identification role:** Older and intermediate firms that cannot buy new equipment enter the used market. Their entry increases competition for the assets young firms normally buy. Young firms lose because they are less able to compete on price, financing, or search.

**Reading cautions:** The tables show relative changes by age group, not every mechanism inside the bargaining process. The paper supports the interpretation with price, trading, and prior-user evidence.

**Economic message:** Secondary markets allocate scarce assets to stronger buyers. This reallocation may be efficient in a narrow willingness-to-pay sense, but it can reduce young-firm investment and potentially affect long-run growth.

## 5.8 Table 9: Prior users of used capital

**Read:** Table 9 compares prior users of used capital with current users during the shortage episode.

**Table 9**  
Prior Users of Used Capital: 2021/22 vs. Normal Times.

|                | (1)<br>$\Delta$ Log Age | (2)<br>$\Delta$ COVID Exposure | (3)<br>$\Delta$ Productivity Growth |
|----------------|-------------------------|--------------------------------|-------------------------------------|
| Year 2021/22   | 0.131***<br>[27.82]     | -0.008**<br>[-1.96]            | 3.330***<br>[16.96]                 |
| Observations   | 499069                  | 197868                         | 43553                               |
| Adjusted $R^2$ | 0.021                   | 0.048                          | 0.114                               |

$t$  statistics in brackets.

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

**Notes:** This table models characteristic differences between the last two users of a specific piece of equipment as a function of time. The dependent variable in Column 1 is the difference between the current and most recent user's log age; in Column 2 it is the difference between the current and most recent user's sector COVID exposure (measured as abnormal employment decline in 2020Q1 (Chodorow-Reich et al., 2022)); in Column 3 it is the difference between the current and most recent user's sector productivity growth rate from the Bureau of Labor Statistics. Fixed effects

Table 9: Prior users and current users of used capital

Table 9 is a reallocation table. It asks whether used assets are moving toward different types of firms or sectors. The finding that current users are older supports the crowding mechanism: older users are moving into markets where younger users previously played a larger role. The productivity-growth comparison suggests another layer: assets may move toward sectors with stronger productivity growth. This makes the welfare interpretation nuanced. Reallocation may raise the current productivity of capital, but it may still harm young firms that could have used those assets for growth.

**Mechanism and identification role:** Table 9 helps explain who wins the competition created by the shortage. The winners are not simply random buyers; they are older and connected to more productive sectors.

**Reading cautions:** Sector-level productivity growth is not the same as firm-level productivity. The table should be read as suggestive evidence about the direction of reallocation, not a complete welfare analysis.

**Economic message:** Used-capital markets reallocate scarcity. The reallocation can be productive-sector oriented, but it also changes firm life-cycle dynamics by reducing access for young firms.

## 6 Short summary

- The paper studies how a shortage of new capital goods changes investment and used-capital markets.
- Figure 1 documents the production-disruption environment.
- Figure 2 and Table 1 show that UCC filings provide asset-level data capable of tracking physical capital over time.
- Table 2 shows substitution away from new equipment and toward older assets.
- Table 3 shows that used assets trade faster, farther, and across less similar users.
- Tables 4–6 use John Deere strike exposure to support the supply-shock interpretation and show corresponding price increases.
- Figure 3 shows why firm age matters: young firms normally use older capital.
- Tables 7–8 show that young firms lose used and total investment during the shortage.

- Table 9 shows that current users during the shortage are older and from sectors with higher productivity growth.
- The central takeaway is that secondary markets both mitigate and transmit investment shocks.

## 7 Conclusion

### 7.1 Core conclusion

A shortage of new physical capital reshapes the secondary market for used capital. The used market partly absorbs the shock because firms can substitute into older assets. But the same market also transmits the shock because firms compete for a fixed stock of used equipment. The resulting reallocation helps some firms keep investing while reducing investment by young firms.

### 7.2 What the paper changes conceptually

The paper changes how we should think about investment when capital supply is fixed. Standard investment analysis often focuses on a firm’s demand for capital or its financing constraints. This paper adds the supply side of the capital-goods market and the secondary market for existing capital. When new capital cannot be produced quickly, the stock of used capital becomes an equilibrium allocation device. The relevant question is not only whether a firm wants to invest, but whether it can acquire an asset when many other firms suddenly want the same used equipment.

### 7.3 Mechanism in one sentence

New capital scarcity raises demand for used substitutes; used-market prices and search intensity rise; stronger or older firms win more of the scarce used assets; young firms lose investment.

### 7.4 Evidence chain

The evidence is cumulative. Figure 1 establishes the timing and macro nature of the shock. Table 2 shows substitution from new to used equipment. Table 3 shows the used market heating up through faster and broader trades. Tables 4–6 validate the supply-side mechanism using John Deere strike exposure and price evidence. Tables 7–8 show the distributional consequence by firm age. Table 9 shows the direction of capital reallocation across prior and current users. Each piece answers a different concern, and together they support the market-clearing mechanism.

### 7.5 Interpretation of efficiency and distribution

The paper does not simply say the reallocation is inefficient. Table 9 suggests that used capital may move toward older users and sectors with higher productivity growth, which could be efficient in a static allocation sense. But this does not remove the distributional concern. Young firms can be crowded out precisely because other users value the assets more or can finance them more easily. Thus, secondary markets can improve contemporaneous allocation while weakening investment opportunities for younger firms.

### 7.6 Policy implications

The findings caution against broad demand-side investment subsidies during fixed-supply shortages. If the supply of new capital is constrained, subsidies can raise competition for the existing stock of capital and increase crowding out in secondary markets. Policies targeted at relaxing supply bottlenecks, improving



access for constrained young firms, or expanding leasing and financing capacity may have different effects from blanket investment incentives. The paper also suggests that policymakers should monitor secondary-market prices and trading patterns, not only new orders and aggregate investment.

## **7.7 Broader implications**

The mechanism applies beyond pandemic-era equipment markets. Any market with durable capital and active resale - vehicles, aircraft, construction machinery, agricultural equipment, computing hardware, and specialized industrial machines - can transmit supply shocks through secondary markets. The paper's logic also connects to macro questions about misallocation, entry, and firm dynamics: shocks to capital supply can affect not only investment quantities but also the identity of investing firms.

## **7.8 Caveats and what not to overinterpret**

The paper does not observe every capital transaction, especially cash purchases outside secured financing and leasing. It also does not directly observe every seller's reason for selling or each buyer's marginal productivity. Firm age is a useful proxy, but it does not fully capture financial constraints, productivity, or growth opportunities. Finally, the welfare implications are not fully pinned down because reallocation toward older or higher-productivity-sector users may have benefits even while young firms are harmed.

## **7.9 Final takeaway**

Secondary markets are not neutral buffers. They make an economy more flexible by reallocating existing capital when new capital is hard to find, but they also determine who bears the shortage. In this paper, the secondary market helps some firms substitute into used equipment while shifting the burden toward young firms. The result is a richer view of investment: capital shortages are mediated not only by financing and demand, but by the equilibrium matching of existing assets across firms.